Lynbrook Math Summer Camp 2026

C4 Probability Applications

Lynbrook Math Club

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C4.1 Lecture

Basic Definitions and Techniques

Probability represents the likelihood of an event happening. It is always defined to be a real number between 0 and 1. An event with probability 1 is certain to happen, while an event with probability 0 is impossible.

Definition C4.1.1. If E represents an event, we denote by $P(E)$ the probability of event E happening.

A high proportion of probability problems involve a set of outcomes. For example, on a coin flip, the set of outcomes is $\{\text{heads}, \text{tails}\}$.

Definition C4.1.2. The **sample space** is the set of all possible outcomes.

Definition C4.1.3. If all outcomes are equally likely, then

$$P(E) = \frac{\text{number of outcomes where } E \text{ occurs}}{\text{total number of outcomes}}.$$

Example 4

A standard 6-sided die is rolled. What is the probability that the number rolled is even?

Definition C4.1.5. Two events are **independent** if one event does not affect the probability of the other event occurring.

Definition C4.1.6. The **complement** of an event E is the event that E does not occur, and is often denoted $\overline{E}$.

Theorem C4.1.7

For any event E, $P(\overline{E}) = 1 - P(E)$.

Definition C4.1.8. Given two events E_1 and E_2, the **union** of E_1 and E_2, denoted $E_1 \cup E_2$, is the event that at least one of E_1 or E_2 occurs. The **intersection** of E_1 and E_2, denoted $E_1 \cap E_2$, is the event that both E_1 and E_2 occur.

Definition C4.1.9. Two events are **disjoint** (or mutually exclusive) if they cannot both occur at the same time; that is, $E_1 \cap E_2 = \emptyset$.

Theorem C4.1.10

For any two events E_1 and E_2,

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2).$$

(You might recognize this as the Principle of Inclusion-Exclusion from yesterday.)

Theorem C4.1.11

If E_1 and E_2 are independent events, then the probability that both events occur equals the product of the probabilities of each event occurring. That is,

$$P(E_1 \cap E_2) = P(E_1) \times P(E_2).$$

Example 12

If you flip a fair coin 4 times, what is the probability of getting exactly 3 heads?

Theorem C4.1.13 (Binomial Probability)

Given n independent trials, each with probability p of success, the probability of obtaining exactly k successes out of n trials is

$$P(k \text{ successes}) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Example 14

Charles and Dan take turns rolling a fair 6-sided die, with Charles going first. The first player to roll a 6 wins. What is the probability that Charles wins?

Geometric Probability

Geometric probability is a technique for solving certain probability problems whose sample space is infinite by converting them into equivalent geometry problems. You are looking for a ratio between the geometric quantities (such as length, area, or volume) to get the probability.

Example 15

A real number x is randomly and uniformly chosen in the interval $[0, 1]$. What is the probability that $x \geq \frac{4}{5}$?

Example 16

Two real numbers x and y are randomly and uniformly chosen in the interval $[0, 1]$. What is the probability that $x + y \geq \frac{4}{5}$?

Conditional Probability

Conditional probability is the probability of an event, given that some other event has already occurred.

Example 17

Sohum flips three fair coins. Given that Sohum did not flip three heads, what is the probability that he flipped three tails?

Definition C4.1.18. Let A and B be events. The notation $P(A \mid B)$ denotes the probability of A occurring, given that B occurred.

Theorem C4.1.19

Let A and B be events. Then $P(A \cap B) = P(A) \times P(B \mid A)$.